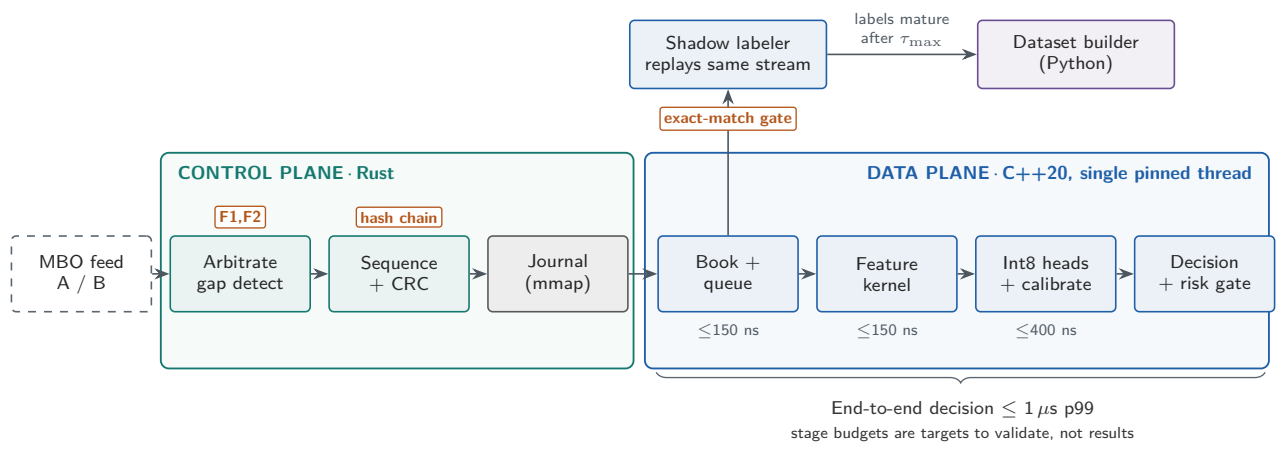
Shadow labeler
replays same stream
labels mature
after $\tau_{\max}$
Dataset builder
(Python)
exact-match gate
CONTROL PLANE · Rust
DATA PLANE · C++20, single pinned thread
F1,F2
hash chain
MBO feed
A / B
Arbitrate
gap detect
Sequence
+ CRC
Journal
(mmap)
Book +
queue
Feature
kernel
Int8 heads
+ calibrate
Decision
+ risk gate
≤150 ns
≤150 ns
≤400 ns
End-to-end decision ≤ 1 μs p99
stage budgets are targets to validate, not results